

1. Administrative & Study Identification

Full Study Title A Phase 1b/2 Open-Label Clinical Study to Evaluate the Safety and Efficacy of MK-6070 and Ifinatumab Deruxtecan (I-DXd) in Participants With Relapsed/Refractory Extensive-Stage Small Cell Lung Cancer **Short Title/Acronym** MK-6070-002 **Protocol Number** MK-6070-002 **NCT Number** NCT06780137 **Posting ID** 915516394246150130 **Sponsor Name** Merck (Merck Sharp & Dohme LLC) **Investigational Product** Gocatamig (MK-6070), Ifinatumab Deruxtecan (I-DXd), and Durvalumab (Imfinzi, ATC: L01FF03) **Phase of Development** Phase 1 / Phase 2 (Phase 1b/2) **Trial Status** Recruiting **Medical Monitor** Merck Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the safety and tolerability (including Adverse Events [AEs], Dose-Limiting Toxicities [DLTs], and study intervention discontinuations due to an AE) of gocatamig alone, I-DXd alone, and gocatamig in combination with I-DXd or durvalumab. * **Secondary Objectives:** To evaluate the efficacy profile, specifically Objective Response Rate (ORR), Duration of Response (DOR), and Progression-Free Survival (PFS).

2.2 Background & Rationale To address the clinical need for novel therapeutic options in patients with extensive-stage small cell lung cancer (ES-SCLC) that has relapsed or is refractory to prior treatments (such as platinum-based chemotherapy). Gocatamig is a DLL3-directed T-cell engager immunotherapy, and ifinatumab deruxtecan is a targeted antibody-drug conjugate. Durvalumab is an additional immunotherapy evaluated in combination.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm (Multiple experimental arms for dose escalation/expansion) * **Blinding:** Open-label * **Randomization:** N/A

3.2 Planned Enrollment & Duration * **Target \$N\$:** 242 evaluable participants * **Number of Sites:** Global (including specific cohorts in Japan and China) * **Estimated Study Start:** 27-Feb-2025 * **Estimated Primary Completion:** 31-Oct-2028

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years. 2. Histologically or cytologically confirmed SCLC that is extensive stage (Stage IV: T any, N any, M1a/b/c) following at least 1 prior line of systemic therapy that included platinum-based chemotherapy. 3. Must be able to provide an archival tumor tissue sample or fresh biopsy tissue sample. 4. Human immunodeficiency virus (HIV) infected participants must have well-controlled HIV on antiretroviral therapy (ART).

4.2 Exclusion Criteria 1. Pleural effusion, pericardial effusion, or ascites requiring recurrent drainage procedures. 2. Untreated or symptomatic brain metastases or a history of leptomeningeal disease. 3. History of (noninfectious) pneumonitis/interstitial lung disease (ILD) requiring steroids, current pneumonitis/ILD, or clinically severe pulmonary compromise. 4. History within 6 months before first dose of coronary/peripheral artery bypass graft, angioplasty, symptomatic CHF (NYHA > class II), uncontrolled arrhythmia, or arterial thrombosis (e.g., stroke or TIA). 5. Active clinically significant infection requiring systemic therapy, or active viral hepatitis (A, B, or C). 6. Receiving chronic systemic steroid therapy (>10 mg daily prednisone equivalent) or chronic immunosuppressive therapy within 7 days prior to the first dose. 7. Active or history of immune deficiency (except well-controlled HIV on ART). 8. Known additional progressing malignancy or history of allogeneic tissue/solid organ transplant. 9. **Part 1 Specific Exclusions:** Radiation therapy to the lung >30 Gy within 6 months; abdominal radiation within 4 weeks; anticancer hormonal treatment (except LHRH) or chloroquine/hydroxychloroquine within 2 weeks; systemic anticancer therapy/investigational agents or antibody-based cancer therapy within 3 weeks; clinically significant corneal disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Part 1:** Assesses the safety, tolerability, and efficacy of gocatamig and I-DXd at doses determined in prior study MK-6070-001 (NCT04471727). * **Part 2:** Assesses the safety and tolerability of gocatamig in participants in Japan and China. * **Part 3:** Assesses the safety, tolerability, and efficacy of gocatamig administered with durvalumab. * **Route of Administration:** Intravenous (IV)/Subcutaneous. * **Duration of Treatment:** Administered until documented disease progression, unacceptable toxicity, or other discontinuation criteria are met.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Antiretroviral therapy (ART) for well-controlled HIV. LHRH for specific hormonal control. * **Prohibited:** Radiotherapy within 2 weeks prior to study intervention, and systemic immunosuppressive therapies/chronic

steroids (>10 mg prednisone/day) within 7 days prior to treatment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor biopsy/tissue collection, Baseline Imaging
Baseline	Day 0	First Dose, Baseline Safety Labs
Interim	Per Cycle	Safety Labs, AE/DLT Monitoring, Efficacy Assessment (Tumor Imaging)
End of Study	Up to 44 Months	Final Vital Signs, Safety Follow-Up, Survival Follow-Up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- 1. Number of Participants Who Experience an Adverse Event (AE):** Untoward medical occurrences temporally associated with the study intervention.
- 2. Number of Participants Who Experience One or More Dose-Limiting Toxicities (DLTs):** Drug-related AEs observed during the DLT evaluation period meeting predefined criteria, graded per NCI CTCAE v5.0 or ASTCT criteria.
- 3. Number of Participants Who Discontinue Study Intervention Due to an AE:** Unfavorable signs/symptoms leading to treatment cessation.

7.2 Secondary Outcomes

- 1. Objective Response Rate (ORR)**
- 2. Duration of Response (DOR)**
- 3. Progression-Free Survival (PFS)**

8. Safety & Adverse Event Reporting

- Adverse Event (AE) Monitoring:** Continuous monitoring collected via site visits and investigator review (graded per NCI CTCAE v5.0). Strict monitoring for ILD/pneumonitis and structural cardiac anomalies.
 - Serious Adverse Events (SAEs):** Evaluated per standard protocol safety guidelines (reporting within 24 hours of awareness).
 - Data Monitoring Committee (DMC):** Safety Review Committee (SRC) to review DLTs during the dose-escalation phases.
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9. Statistical Analysis Plan (SAP) Summary

- Analysis Populations:** Safety Analysis Set (all participants receiving ≥ 1 dose of study drug); Efficacy Analysis Set (ITT/Per-Protocol equivalent populations).
- Sample Size Justification:** Target enrollment is $N=242$, powered to evaluate standard Phase 1 safety/tolerability

dose-limiting toxicity rules and Phase 2 objective response rate targets across Parts 1, 2, and 3.

- **Handling of Missing Data:** Standard oncology conventions (e.g., censoring for PFS at the last evaluable tumor assessment date).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine onsite and remote risk-based monitoring visits per Sponsor clinical plan.
 - **Audit Trail:** Secure Electronic Data Capture (EDC) system with full query tracking and eCRF locking mechanisms.
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11. Study Identifiers & Governance

- **Primary ID:** {{MK-6070-002}}
 - **Registry Number:** {{NCT06780137}}
 - **Posting ID:** {{915516394246150130}}
 - **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC}}
 - **Study Title:** {{A Study to Evaluate the Safety and Efficacy of Gocatumig (MK-6070) and Ifinatumab Deruxtecan (I-DXd) in Participants With Relapsed/Refractory Extensive-Stage Small Cell Lung Cancer (MK-6070-002)}}
 - **Official Regulatory Title:** {{A Phase 1b/2 Open-Label Clinical Study to Evaluate the Safety and Efficacy of MK-6070 and Ifinatumab Deruxtecan (I-DXd) in Participants With Relapsed/Refractory Extensive-Stage Small Cell Lung Cancer}}
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12. Clinical Framework (Oncology/SCLC Specific)

- **Primary Condition:** {{Relapsed/Refractory Extensive-Stage Small Cell Lung Cancer}}
 - **Diagnosis Threshold:** {{Histologically/cytologically confirmed Stage IV SCLC post-platinum chemotherapy}}
 - **Medical Methodology:** {{Immunotherapy (DLL3-directed T-cell engager, Durvalumab) and targeted Antibody-Drug Conjugate therapy}}
 - **Optimization Target:** {{Maximum Tolerated Dose / Recommended Phase 2 Dose, AE management, and Objective Response Rate}}
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13. Study Procedures & Methodology

- **Management Pathway:** {{Standard oncology clinical pathway for relapsed/refractory SCLC}}

- **Education Protocol (HCP):** {{Site Initiation Visits and Investigator Brochure (IB) toxicity management training}}
- **Education Protocol (Patient):** {{Informed Consent review and ILD/pneumonitis safety symptom monitoring}}
- **Quality Audit Mechanism:** {{Clinical site monitoring and EDC validation}}

14. Observation & Follow-Up Schedule

- **Total Duration:** {{Up to approximately 44 months}}
- **Onsite Visit Cadence:** {{Every study cycle for dosing and assessments}}
- **Remote Monitoring Frequency:** {{Continuous safety monitoring per protocol}}
- **Checklist Review Interval:** {{Tumor imaging evaluations every 6-8 weeks}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				